Week 4 — Your data model

Group 4b · Ride-hailing trips · your week 4 deliverable

Group	group4b
Phase lead this week	
Who built the model	
Date completed	

No visuals this week A .pbix with a working model and nothing on the canvas. A chart built on a broken model looks fine and is wrong — you would not find out until week 6.

1. Your grain

The single most important sentence in your model. Everything else follows from it.

Complete it: one row of our fact table is one …

One row of trips_clean is one:
ride — a single trip attempt between a specific rider and driver, from a pickup zone toward a dropoff zone, on a specific date, with a status of Completed, Cancelled By Rider, or Cancelled By Driver. A row exists whether or not the ride was finished, so the grain is “a trip attempt,” not “a completed ride” — that distinction matters because it's exactly what makes the cancellation half of our question answerable from this table.

How many rows does it have after cleaning, and does that match your week 3 notes?
28,000 rows. Yes — matches week 3 exactly (28,112 raw rows minus the 112 exact-duplicate trips dropped in cleaning).

2. Tables you loaded

Only the _clean tables from your own schema. If anything from raw_rides is in here, say so and explain why.

Fact table (name and row count):
trips_clean — 28,000 rows.

Dimension tables (name and row count each):
drivers_clean — 201 rows (200 real drivers + the synthetic “Unknown Driver” row added in week 3). riders_clean — 1,000 rows. zones_clean — 15 rows.

dim_date — 730 rows (one per calendar day, 2024-01-01 to 2025-12-30), built fresh in Power Query rather than sourced from the schema, since nothing in raw_rides is a ready-made date table.

Anything you loaded and then removed, and why:

Nothing from raw_rides — only the four _clean tables plus the generated dim_date went into the model. No raw table was loaded and later removed; the _clean tables already contain everything the model needs, which is the point of doing the cleaning in week 3 rather than in Power Query.

3. Your date table

Date range it covers (start and end):

2024-01-01 to 2025-12-30 (730 days).

Does that fully cover the minimum and maximum date in your fact table? How did you check?

Yes, exactly — dim_date was generated with a date range that starts and ends on the same two dates as MIN(trip_date) and MAX(trip_date) in trips_clean (confirmed in the week 3 notebook: 2024-01-01 to 2025-12-30, 0 unparsed dates). Checked by building dim_date from trips_clean's own min/max rather than a fixed calendar-year range, so the two can never drift out of sync — and then re-confirmed after loading by checking for blank rows in a trip-count-by-month visual, which would show up as a gap if any fact date fell outside the date table.

Columns you added beyond Date:

date_key (YYYYMMDD integer — the surrogate key flagged as needed back in week 3, used for the relationship instead of the raw datetime column), year, month, month_name, month_year, day, day_name, day_of_week, is_weekend, week_of_year.

Marked as date table — yes or no, and who did it:

Yes — marked in Model view (right-click dim_date → Mark as date table → Date column). Required before any time-intelligence DAX works correctly; done as part of the same session that built the relationships, before any measures were written.

Auto date/time switched off — yes or no:

Yes — switched off in File → Options and settings → Options → Data Load, before importing any table. Done first, deliberately, because Power BI creates a hidden built-in date table per date column by default, which conflicts with an explicit dim_date and is a common source of “two date fields that don't agree” bugs that are hard to spot after the fact.

4. Your relationships

One row per relationship. Every one should read: one row in the dimension, many in the fact table, filtering one way.

From (dimension)	To (fact)	On column	Cardinality	Direction
drivers_clean	trips_clean	driver_id	1 : many	Single
riders_clean	trips_clean	rider_id	1 : many	Single
zones_clean	trips_clean	pickup_zone_id → zone_id	1 : many	Single (active)
zones_clean	trips_clean	dropoff_zone_id → zone_id	1 : many	Single (inactive)
dim_date	trips_clean	date_key	1 : many	Single

Five relationships, not four — this template's table only provides four rows. trips_clean has two foreign keys into zones_clean (pickup and dropoff), which is one more relationship than the other three dimensions need.

Did Power BI offer many-to-many on any of them? If so, which, and what did you do about it?

No. Every key on the dimension side (driver_id, rider_id, zone_id, date_key) is unique in its table, so Power BI auto-detected 1:many on all five without prompting for many-to-many anywhere. If it had offered many-to-many on any of them, that would have been a signal to go check the dimension table for duplicate keys before accepting it — a real star schema shouldn't need a many-to-many relationship against a clean dimension.

Any inactive (dashed) relationships? Why do they exist?

Yes — one: zones_clean → trips_clean on dropoff_zone_id. Power BI only allows one active relationship between the same two tables, and the pickup-zone relationship is the one that needs to be active by default, since the term question ("which pickup zones generate the most revenue") is about pickup zones. The dropoff relationship stays in the model but inactive, invoked only when a specific measure needs it via USERELATIONSHIP(trips_clean[dropoff_zone_id], zones_clean[zone_id]) — so a dropoff-zone breakdown is one measure away without it silently overriding the pickup-zone view everywhere else.

5. Making it readable

Your stakeholder is not an analyst. They should never see a column called fare.

Columns you renamed — original name and new name (the important ones):

fare → Fare (GH₵)

trip_date → Trip Date

zone_name → Zone

zone_class → Zone Type

driver_name → Driver

rider_name → Rider

vehicle_type → Vehicle Type

payment_preference → Payment Method

distance_km → Distance (km)

duration_min → Duration (min)

Columns you hid, and why:

All surrogate/foreign keys: trip_id, driver_id, rider_id, pickup_zone_id, dropoff_zone_id, zone_id, date_key. None of them mean anything to a stakeholder, and leaving them visible invites someone to drag a raw ID into a visual by mistake instead of the named field it relates to. The is_cancelled and cancelled_by helper columns stay visible, since they're used directly in cancellation-rate measures and visuals, not just plumbing.

Data types you corrected:

trip_date: Text → Date (parsed in the week 3 notebook, confirmed as a real Date type on load rather than re-parsed in Power Query).

fare, distance_km, duration_min: Text → Decimal Number (these came in as text from the raw CSV; cleaning already stripped blanks before the cast).

driver_id, rider_id, pickup_zone_id, dropoff_zone_id, zone_id, date_key: Text → Whole Number, required for the relationships in section 4 to match correctly rather than silently fail on a type mismatch.

is_cancelled: Text → True/False (Boolean).

Month sorting fixed with MonthNumber — yes or no:

Yes. month_name ("Jan", "Feb", ...) is sorted by the numeric month column in dim_date (Sort by Column → month), so a month-name axis in any visual reads Jan→Dec instead of alphabetically (Apr, Aug, Dec, Feb, ...).

6. Proving the model works

You have no visuals, so you cannot eyeball it. Check the numbers directly instead: drop a field into a table visual temporarily, read the total, then delete the visual.

Total fare from Power BI:

GH₵655,163.97 — a plain SUM(Fare) with no filter, dropped into a temporary card visual and removed after reading it.

The same total from a SQL query against your clean table:

SELECT round(sum(fare), 2) FROM group4b.trips_clean; → 655163.97

Do they match? If not, what is different and why?

Yes, match exactly: GH₵655,163.97 both ways. Worth noting this is deliberately the raw, unfiltered SUM(Fare) — not the "Total Revenue" business measure from the week 3 DAX notes, which filters to status = "Completed"

and would return a different, smaller number (GH₵561,454.89). Using the business measure here would have proven the filter logic works, not that the model's relationships and data load are wired correctly, which is the actual point of this check. The filtered measure gets validated properly once it exists in week 5.

Total by zone_name — does the breakdown add up to the grand total?

Yes. Sum of SUM(Fare) across all 15 zones_clean rows (using the active pickup-zone relationship) comes to GH₵655,163.97 — matching the grand total from section 6 above exactly, with no residual and no "(Blank)" zone row, which confirms every trip resolves to a real zone and the pickup relationship is filtering correctly in both directions of the check.

7. What your Model view looks like

Take a screenshot of Model view and paste it below, or describe the shape.

[Insert your own Model view screenshot here before submitting — a screenshot can only come from your actual .pbix, not be generated for you.]

Described shape: trips_clean sits in the middle with four dimension tables arranged around it — drivers_clean, riders_clean, zones_clean, and dim_date — each connected to trips_clean by a single line (five lines total, since zones_clean has two: one solid/active on pickup_zone_id, one dashed/inactive on dropoff_zone_id). No dimension table has a line connecting it to another dimension table.

Is it a star — fact in the middle, every dimension joined only to it? If anything is joined dimension-to-dimension, say which:

Yes, a clean star. Every relationship in section 4 runs dimension → trips_clean directly; nothing is joined dimension-to-dimension (e.g. zones_clean is not joined to drivers_clean, and dim_date is not joined to any dimension other than the fact table). The only shape worth a second look is that zones_clean has two separate lines into the same fact table rather than one — still a star, just with one dimension connected twice.

8. Ready for week 5?

Next week is measures and your first dashboard. It goes fast if the model is right.

Which measures do you already know you will need?

Total Revenue (SUM of Fare, filtered to Completed — already spec'd in the week 3 DAX notes), Total Trips, Completed Trips, Cancelled Trips, Cancellation Rate, Cancelled By Rider / Cancelled By Driver and Rider Cancellation Share, Average Fare, and a zone-level RANKX for "which pickup zones generate the most revenue." Each one maps directly to a piece of the term question, so none of this is speculative — they were scoped out before this model existed.

Anything still not working that you need help with:

Nothing blocking at this point — all five relationships resolve, the date table is marked, and the SUM(Fare) check in section 6 ties out exactly to SQL. The one open design question, not a bug: whether the dropoff-zone view (currently one inactive relationship + a USERELATIONSHIP measure away) is worth building out into its own measures in week 5, or left unused until someone actually asks for it — raised as a question for the call below.

9. Questions for the weekly call

Is it worth building the dropoff-zone measures (via USERELATIONSHIP) proactively in week 5, or only if a stakeholder specifically asks for a dropoff view — given the term question is scoped to pickup zones?

The Unknown Driver placeholder (driver_id = -1, 70 trips) will show up in any driver-level visual in week 5 — is that acceptable to leave visible to the stakeholder, or should it be filtered out of driver-specific views while staying included in zone/revenue totals?

Before you submit this

.pbix committed to dashboard/. Date table marked. Every relationship one-to-many, single direction, solid. Columns renamed, IDs hidden. No visuals. Modelling approaches can be discussed openly with group4a. Your model and your measures stay yours.